MINOR PROJECT REPORT

STUDENT PERFORMANCE ANALYSIS AND PREDICTION SYSTEM

Using Python and Machine Learning

Submitted by	Manish Kumar
Roll Number	_____
Course	AI/ML with Python Training Program
Batch	September 2026
College / Institute	Shri Shankaracharya Professional University, Bhilai
Project Guide	Pratham Saxena

Academic Project Submission • 2026

DECLARATION

I hereby declare that the project entitled **“Student Performance Analysis and Prediction System”** has been prepared by me as a minor project using Python and machine-learning techniques. The work presented in this report is based on the implementation carried out in the project notebook. The dataset used for this demonstration is a sample/synthetic dataset; therefore, the results should not be interpreted as findings about real students.

ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, **Mr. Pratham Saxena**, and **Shri Shankaracharya Professional University, Bhilai**, for their guidance and support. I also acknowledge the learning resources and open-source Python libraries that helped me complete this project.

ABSTRACT

This project presents a Student Performance Analysis and Prediction System developed with Python. It demonstrates a basic machine-learning workflow: loading a student dataset, cleaning the data, exploring patterns through visualizations, training a Linear Regression model, and evaluating model performance. Study hours per day, attendance percentage, and previous examination marks are used as input features to estimate final examination marks. The model is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 score, and five-fold cross-validation.

1. INTRODUCTION

Student performance can be influenced by several factors, including study habits, attendance, and prior academic performance. Data analysis and machine learning can help explore relationships among these factors and build simple predictive models. This project demonstrates the end-to-end process on a small sample dataset.

2. PROBLEM STATEMENT

The objective is to analyze a structured student dataset and build a regression model that estimates final examination marks from selected academic indicators. The project is educational and exploratory; it is not intended to make decisions about real students.

3. OBJECTIVES

• Load and inspect a CSV dataset using Pandas.

• Identify and handle duplicate records and missing numerical values.

• Explore the dataset using descriptive statistics and visualizations.

• Train a Linear Regression model using selected input features.

• Evaluate predictions using MAE, RMSE, R^2, and cross-validation.

• Document the workflow, results, limitations, and possible future improvements.

4. TOOLS AND TECHNOLOGIES

Programming language: Python
Environment: Google Colab
Libraries: Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn
Model: Linear Regression

5. DATASET DESCRIPTION

The dataset contains 301 records before duplicate removal and five columns: student_id, study_hours_per_day, attendance_percent, previous_exam_marks, and final_exam_marks. After removing duplicate records, the dataset contained 300 rows. The three academic indicators were used as model inputs, while final_exam_marks was used as the target variable.

6. METHODOLOGY

6.1 Data loading and inspection

The CSV file was loaded into a Pandas DataFrame. The first rows, dataset dimensions, column information, descriptive statistics, and missing-value counts were inspected.

6.2 Data cleaning

Duplicate rows were removed. Missing values in numerical columns, if present, were filled using the mean of the respective column. The cleaned dataset contained 300 rows and five columns.

6.3 Exploratory data analysis

A histogram with a density curve was created to examine the distribution of final examination marks. A correlation heatmap was generated for the numerical columns to visualize pairwise relationships.

6.4 Model development

The input variables were study_hours_per_day, attendance_percent, and previous_exam_marks. The target variable was final_exam_marks. The data was divided into training and testing subsets using an 80:20 split with random_state=42. A Linear Regression model was fitted on the training data.

6.5 Model evaluation

Predictions were generated for the test set. MAE, RMSE, and R^2 were calculated. Five-fold cross-validation using R^2 as the scoring metric was also performed on the dataset.

7. RESULTS

Evaluation metric	Observed result
Mean Absolute Error (MAE)	6.8797
Root Mean Squared Error (RMSE)	8.2824
Test R^2 score	0.5000
5-fold cross-validation R^2 scores	0.5603, 0.4895, 0.4968, 0.5302, 0.6408
Average cross-validation R^2	0.5435

The average cross-validation R^2 score was 0.5435. This indicates a moderate fit for this sample dataset. R^2 is not the same as percentage prediction accuracy. The results depend on the sample data and should not be generalized to real educational settings.

8. CONCLUSION

The Student Performance Analysis and Prediction System was implemented successfully using Python and Linear Regression. The project covered data loading, cleaning, exploratory data analysis, visualization, model training, evaluation, and cross-validation. The model achieved a test R^2 score of approximately 0.5000 and an average five-fold cross-validation R^2 score of 0.5435 on the sample dataset. The project demonstrates a basic machine-learning workflow and provides a foundation for further study.

9. LIMITATIONS AND FUTURE SCOPE

• The dataset is synthetic/sample data and may not represent real student populations.

• Only three input features were used; other relevant factors were not included.

• Linear Regression may not capture complex non-linear relationships.

• Future work may use a larger, validated real-world dataset with appropriate privacy safeguards.

• Other regression algorithms, feature engineering, and hyperparameter tuning may be explored and compared.

10. REFERENCES

• Pandas documentation: https://pandas.pydata.org/docs/

• Scikit-learn documentation: https://scikit-learn.org/stable/

• Matplotlib documentation: https://matplotlib.org/stable/

• Seaborn documentation: https://seaborn.pydata.org/

11. APPENDIX – PROJECT EVIDENCE

Add genuine screenshots from your Google Colab notebook in this section or as separate appendix pages before final submission. Recommended evidence: (1) dataset preview, (2) data-cleaning output, (3) final-marks histogram, (4) correlation heatmap, (5) model training confirmation, (6) evaluation metrics, (7) actual-versus-predicted graph, and (8) cross-validation scores. Do not insert fabricated screenshots.

Student signature: _____ **Date:** _____

Guide signature: _____